SE-4096
Data Science for Software Engineering

Muhammad Yaseen Khan, PhD

muhammad.yaseen@nu.edu.pk

About Me

- Background in Software Engineering, Data Science, and AI
- Research Focus: Software Engineering, NLP, Psychology
- Worked at/for: Microsoft, Alibaba, Alexa Translations
- Taught at: KIET, MAJU, KSBL, and Habib University
- Thursday, 8:00 a.m. - 10:00 a.m. at department of AI & DS

Google Scholar | Linkedin

About Course

This course introduces data-driven techniques for analyzing and improving software systems using real software engineering data such as source code, repositories, issues, and logs.

Students learn to apply statistical and machine learning methods to support evidence-based decisions related to software quality, maintenance, and productivity through hands-on analysis and projects.

Why to have this course

- Evidence-based decision making
- Improved software quality
- Better maintenance and evolution
- Productivity and team insights
- Risk reduction in software projects
- Alignment with modern industry practices
- Effective use of large-scale software data
- Et cetera.

Course Objectives

By the end of this course, students will be able to:

- To enable students to **apply data science workflows and analytical techniques** to real software engineering problems and artifacts.
- To develop the ability to **collect, preprocess, and analyze software engineering data** from repositories, issue trackers, and operational systems.
- To equip students with skills to **build and evaluate predictive models** for software quality, maintenance, and process improvement.
- To foster **evidence-based decision making** in software engineering using empirical data and visualization.

Course Demands

- Understanding of Probability and Statistics, Data Structures, Software Engineering, Software Testing, Software Project Management
- Active participation
- Bringing the quantified/classified answers to the SE practices.

Learning Approach

- Brief introduction to the topics through lectures with real-life analogies
- Data-centric thinking, raising questions on/of/to the data
- Statistical computations, Programming (Python), Analysis of the visualization
- Analysing the SoTA datasets, papers, reproducing them

Content Overview

- Introduction to course
- DS Workflow & Software Data Sources
- Data Collection & Preprocessing
- Exploratory Data Analysis (EDA)
- Software Metrics & Feature Engineering
- Machine Learning Fundamentals
- Defect Prediction Models
- Effort Estimation & Cost Prediction
- Software Maintenance & Evolution Analytics
- Developer & Team Analytics
- Experiment Reporting & Future Trends

Reading Material

- Few of them are uploaded on GCR.
- Research Papers.
- Official none, drive your own as you want.

Tentative Marks Distribution

- Assignment/Quizzes: 15 %
- Sessional Exams: 30 %
- Final: 30 %
- Projects + Term Paper: 25 %

Deadlines are always final.

Honesty prevails over everything.

Introduction to **Data Science in Software Engineering**

Data *vs.* Information

- Data is **raw facts** and **observations**
- Data by itself has no meaning
- **Information:** processed and meaningful data

Examples:

  - Number of commits: 25 (data)
  - Bugs reported: 3 (data)
  - The average number of commits/day is 5 (info.)

Takeaway:

Data is like raw ingredients (flour, sugar, eggs).
You can’t eat them directly — you must process them

Is the word *data* plural or singular?

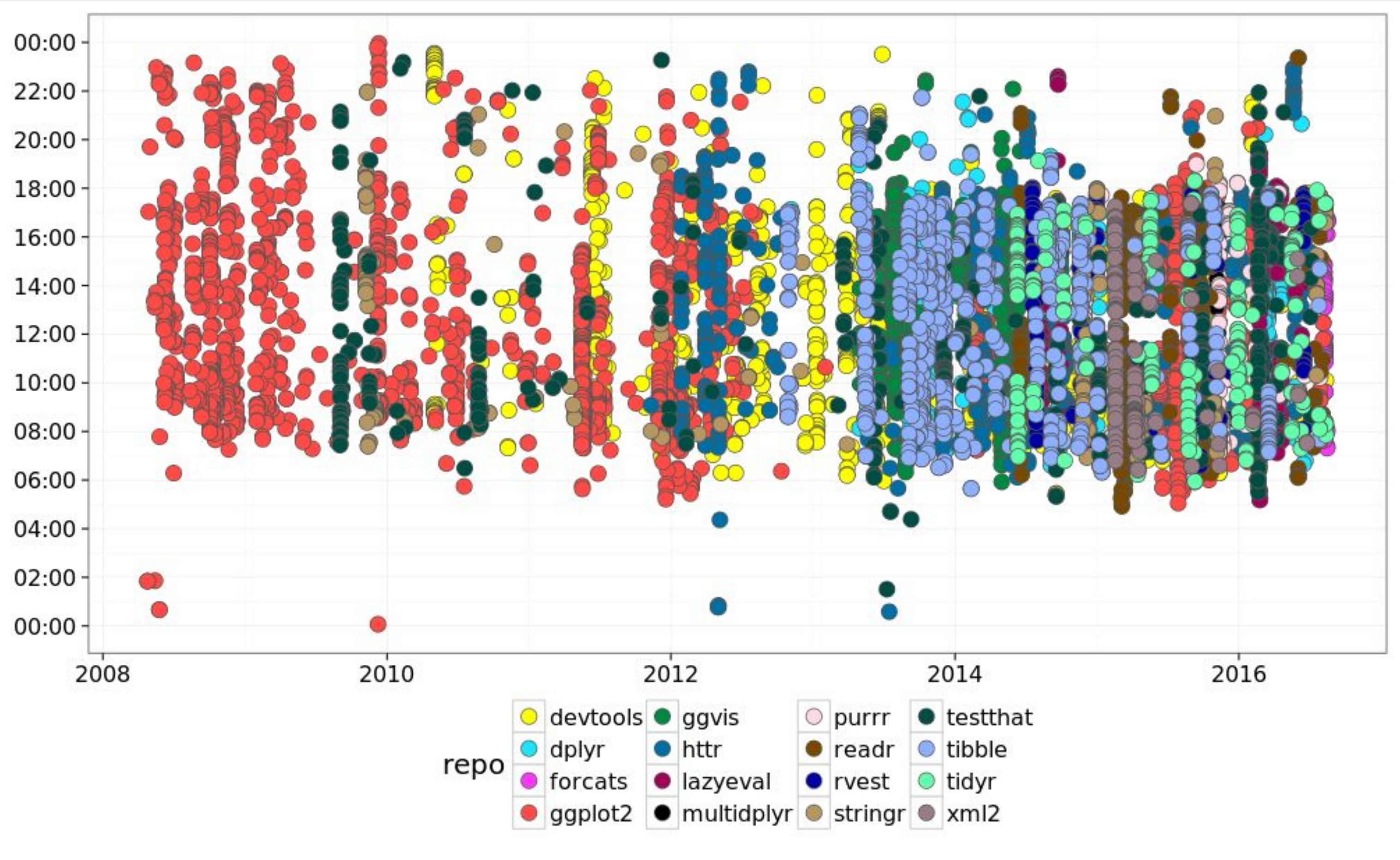
What is Data Analytics?

- Focuses on:
  - analyzing existing data
  - finding patterns
  - answering “what happened … ?” … “which is … ?”
- Mostly:
  - Charts
  - Reports
  - Dashboards

Example:

  - Which file had the most bugs last month?

Given the time of git-commit on repo; how would you build a visual report ?



https://deanattali.com/blog/visualize-git-commits-time/

What is Data Science?

- Goes *beyond* analytics
- Includes:
  - Statistics
  - machine learning
  - prediction
- Answers:
  - “What will happen?”
  - “Why did it happen?”

Takeaway:

- Analytics = looking in the rear-view mirror
- Data Science = predicting what’s ahead

Data Analytics vs Data Science

Data Analytics	Data Science
Past-focused	Future-focused
Descriptive	Predictive
Reports	Models
“What happened?”	“What will happen?”

Both are important, but data science helps planning.

Given the visualization above, can you answer the idle time of programmer for next year?

What is Data Science for Software Engineering?

- Applying data science to software engineering problems
- Uses:
  - Code
  - Commits
  - Bugs
  - logs
- Goal:
  - better quality
  - Automated risk prediction / less risk
  - Automated priority classification / smarter decisions

Real-Life SE Problems Solved Using Data Science

- What would be the effort required for building the SW
- What would be priority and severity of the bug when CI/CD fails for a certain feature push?
- Which commit may break the system?
- Which module needs refactoring?
- Which developer workload is too high?
- Et cetera.

Software Engineering as a Data-Rich Discipline

- Every development activity creates data:
    - writing code
    - fixing bugs
    - running tests
    - Many digital footprints on Git/VCS
- Tools automatically record everything

Types of Software Engineering Data

- Source code (files, classes)
- Commits (Git history)
- Bugs and issues
- Developers' activity
- Test results
- Logs and runtime data

How Do We Use SE Data?

- Code data ➔ predict defects
- Commit data ➔ risky changes
- Bugs data ➔ effort estimation
- Logs ➔ failure detection
- Developer data ➔ productivity analysis

Summary and Takeaway

- Data is everywhere in software engineering
- Data science helps:
  - understand
  - predict
  - improve software
- This course will teach you:
  - how to use software data
  - to make smart engineering decisions

Lab:

- Introduction to
  - Python, Anaconda distribution, and modules for DS:
    - Pandas
    - Matplotlib
    - scikit-learn
  - Git API